

Pavitha M

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EDUCATION

Rajalakshmi Engineering College, Chennai, India

Cumulative GPA: 8.64

B.Tech in Computer Science and Business Systems

TECHNICAL SKILLS

- **Languages:** Python, C, Java, C++
- **Frameworks and Libraries:** Flask, Celery, FastAPI
- **Databases:** MySQL, Redis
- **Tools & Platforms:** Git, Github, AWS(basics)

PROJECTS

<https://github.com/Pavitha-07>

1. Automated Codebase Documentation Generator

Python • Docker • Redis • Celery

- Built an autonomous AI agent pipeline that clones GitHub repositories, analyzes source files and dependencies, and auto-generates Markdown documentation, README files, and usage guides.
- Designed a queue-based async backend using *Celery* and *Redis* to handle long-running repository analysis tasks without blocking the user-facing interface.
- Containerized the entire system with Docker to ensure reproducible environments and production-ready deployment.
- Developed a frontend interface for submitting GitHub URLs and triggering automated documentation generation end-to-end.

2. LLM Model Router

Python • FastAPI

- Built a model routing system that dynamically selects the most suitable AI model based on request characteristics such as latency, cost, and task requirements.
- Implemented FastAPI-based REST endpoints to route incoming requests and manage model selection logic.
- Designed routing strategies to optimize performance and cost efficiency across multiple AI models.

3. CropDoc — AI-Powered Agricultural Assistance Platform

React • Tailwind CSS • Python(FastAPI) • TensorFlow | Team project

- Built a full-stack AgriTech platform helping farmers make data-driven decisions through AI-powered crop recommendations, pest prediction, and real-time environmental insights.
- Developed a React + Tailwind CSS frontend and FastAPI backend to serve TensorFlow-based models for pest detection and fertilizer recommendations.
- Integrated location-specific weather updates, crop calendars, market price tracking, and a multilingual AI voice bot for accessibility in low-literacy farming communities.

CERTIFICATIONS & ACTIVITIES

- Accenture Software Engineering Virtual Job Simulation (Forage) - completed a simulated SDLC workflow covering architecture, coding, and testing.
- Technical Team Member, IEEE TEMS Society — contributed to organizing technical events, workshops, and knowledge-sharing sessions for the student chapter
- Hackathons: Winner (Problem Statement Category) at Internal Hackathon(SIH); National Finalist at AI Ignite; Finalist in two additional national/regional hackathons across AI and full-stack domains.